

Finite-Resolution-Relaxed Ref-GH Puncture-Exponent Controller

Partial Qualification Report and Aurora PVC Blocker

AthenaK vacuum reference-frame first-order GH

26–27 August 2026

Branch	codex/ref-gh-q-relaxed-controller-20260826
Exact base	70e1579c2ed117e538e79b4cdf9b461ec58330e8
Implementation	e403baf03775a04d14c7fa8d50b31a0d44c50c24
PVC layout correction	a3d9818220a509d9b749373f315a2e82ddb44902

1 Executive status

The analytic estimator, finite- h oracle, q-controlled provider, physical and gauge reprojection, scalar controller, stage ordering, and controller-off static-reference ladder are implemented. Nine local cases reach $t = 0.1M$ at $h = M/16, M/24, M/32$. In the regular annulus $0.25M \leq r < 0.375M$, the $q = 0.9$ and $q = 1.1$ representations approach the same physical stationary trumpet at roughly third-to-fourth order; $q = 1$ remains at roundoff.

The prescribed moving-reference and closed-loop gates are **FAIL**. Aurora jobs 8785680 and 8785718 pass the one-tile analytic/cache gate, then encounter the same eight-rank Level Zero **NotPresent** GPU write fault in the first evolved cycle. The second job follows one equation-preserving boundary refactor on a different node. No third micro-experiment was launched. This document does *not* claim finite-resolution control established, closed-loop relaxation, moving-reference convergence, restart qualification, PVC evolved qualification, or production readiness.

2 Estimator and finite-resolution target

At each active Cartesian cell, with $X^i = x^i - x_p^i$,

$$q_{\text{loc}} = -\frac{1}{6}X^k\gamma^{ij}\partial_k\gamma_{ij} = -\frac{1}{6}X^k\partial_k\ln\det\gamma. \quad (1)$$

The physical metric derivative is reconstructed from Ψ , Φ , the current reference frame/coframe, and analytic reference jets. The production controller performs no spatial finite difference, ray interpolation, radial binning, or interpolation to $r = 0$.

For actual cells in $2h \leq r < 8h$,

$$w_i = \left(\frac{2h}{r_i}\right)^3, \quad q_{\text{est}} = \frac{\sum_i w_i q_{\text{loc},i}}{\sum_i w_i}. \quad (2)$$

Variance, effective sample size, extrema, and count are recorded. The exact finite-grid target $q_T^{\text{analytic}}(h)$ applies the identical cell set and weights to the provider-derived finite-radius exponent. The controller is not told that the continuum limit is one.

The independent relative diagnostic is

$$\epsilon_G = -\frac{1}{6}X^k(G^{-1})^{IJ}\bar{\theta}^K{}_k\Phi_{KIJ}, \quad q_{\text{phys,loc}} = q_{\text{ref,loc}} + \epsilon_G. \quad (3)$$

The identity passes analytic/random-state tests; ϵ_G is recorded and is not forced to zero.

2.1 Puncture-stencil exclusion

Every primary field, constraint, common-ADM, exponent, gauge, and cache-oracle sample is discarded whenever its complete FD support can overlap the puncture. The conservative support box uses each block direction’s spacing. An order $2p$ centered derivative has radius p ; enabled matching KO adds one cell. The common ADM mask uses the larger of its diagnostic and evolution footprints. The anisotropic-spacing oracle passes and retains 2,112 of 2,144 estimator cells. No post-hoc mask was introduced after inspecting the runs.

3 Static analytic gates

Minkowski gives $q_{\text{loc}} = 0$ to roundoff. For $\psi = 1 + M/(2r)$, the exact wormhole result

$$q_{\text{exact}}(r) = \frac{M}{r + M/2} \quad (4)$$

is reproduced at selected cells and approaches two under refinement. The trumpet comparator uses the same interpolated profile and analytic derivative as the provider. Its weighted values are

h/M	$q_T^{\text{analytic}}(h)$
1/16	0.8437263470
1/24	0.8938418434
1/32	0.9202374342

and trend toward one. The same- r/h direct-FD estimate is retained only as a diagnostic of scale-invariant singular-stencil bias. Its hard gate was removed; a fixed-coordinate annulus FD check converges.

4 q-controlled reference and reprojection

Only the exact trumpet spatial Cholesky factor changes:

$$W(r) = e^{-(r/R_G)^2}, \quad L(q) = L_T \exp[-(q-1)W(r) \ln(r/M)], \quad R_G = 3M. \quad (5)$$

Lapse, shift, finite-radius profiles, and R_G remain fixed. At $q = 1$ the provider agrees with L_T to binary64 accuracy. For $q \neq 1$, the same physical stationary coordinate metric and derivatives are projected into the current tetrad. The resulting Π is generally nonzero because the physical shift advects the relative metric. Reconstruction errors are 4.44×10^{-16} for the metric and 9.86×10^{-16} for its derivative; the smallest deliberately nontrivial $|\Pi|$ is 0.125187. The ordinary-GH $\hat{H}$ and stationary improved-driver θ are likewise projected and subtract the matching current-reference baseline. Their source-unit errors are 1.11×10^{-16} and 2.22×10^{-16} .

Nonperiodic boundaries project this same physical metric and gauge state at the stage values $(q, \dot{q}, \ddot{q})$. They never reuse $\Psi = \eta, \Pi = \Phi = 0$ for a mismatched reference.

5 Controller and RK coupling

The replicated state $(q, \dot{q})$ obeys

$$\dot{q} = q_{\text{dot}}, \quad (6)$$

$$\dot{q}_{\text{dot}} = -2\zeta\omega_q q_{\text{dot}} + \omega_q^2(q_{\text{est}} - q), \quad (7)$$

with a smooth acceleration limiter. Defaults are $\zeta = 1$, $\omega_q M = 0.5$, $q \in [0.5, 2.5]$, and $|\dot{q}|M \leq 0.25$. State-bound crossing fails the run; there is no clipping-and-continuation path.

The PDE and controller use identical RK stages. A stage copies the PDE and controller state, builds a lightweight current-q spatial snapshot, measures q_{est} , computes $\ddot{q}$, builds the full two-jet cache with the same generation,

computes the Ref-GH/gauge RHS, and performs the common update. Generation assertions reject stale reference use. Restart headers persist the controller state, generation, frozen state, and estimator diagnostics.

The manufactured scalar histories verify sign, critical damping, slowly varying response, exact restart continuation, prescribed C2 pulses, and absence of clipping. Critical-history error is approximately 1.03×10^{-15} ; the Aurora/SYCL prescribed oracle error is 3.02×10^{-9} .

6 Controller-off static ladder

All cases use $[-0.5M, 0.5M]^3$, exact projected boundaries, CFL 0.05, RK4, fourth-order FD, $\gamma_0 = \gamma_2 = 1$, no KO, and the complete puncture mask. Every case reaches $t = 0.1M$ finite without intervention.

q	h/M	metric	native constraint	lapse	shift
0.9	1/16	1.5568e-3	1.5670e-2	6.4446e-7	9.8937e-7
0.9	1/24	3.6211e-4	4.1008e-3	1.7590e-7	9.5102e-8
0.9	1/32	1.2322e-4	1.4604e-3	3.8504e-8	1.4421e-8
1.1	1/16	2.0136e-3	1.5609e-2	7.9275e-7	1.0139e-6
1.1	1/24	4.8115e-4	4.0819e-3	2.1548e-7	9.9415e-8
1.1	1/32	1.6607e-4	1.4529e-3	6.9710e-8	2.0521e-8

These are puncture-safe Linf values in $0.25M \leq r < 0.375M$. Metric orders are 3.60, 3.75 for $q = 0.9$ and 3.53, 3.70 for $q = 1.1$; constraint orders are 3.31, 3.59 for both signs. The $q = 1$ errors lie at roughly 10^{-15} – 10^{-12} , so no convergence order is assigned.

The innermost $0.125M \leq r < 0.25M$ metric Linf is **FAIL** monotone: $q = 0.9$ gives 4.8404e-2, 6.5666e-3, 1.1201e-2, and $q = 1.1$ gives 6.6575e-2, 1.2940e-2, 1.5590e-2. This negative result is retained.

7 Prescribed moving-q evidence

A local full-output $1 \rightarrow 0.9 \rightarrow 1$ startup reaches $t = 0.05M$ finite. Trajectory errors are about 10^{-15} in q and 2.4×10^{-14} in $\dot{q}$. Final metric, lapse, and shift errors are 8.27×10^{-11} , 3.18×10^{-13} , and 1.24×10^{-12} ; the native constraint Linf is 3.47×10^{-9} . This is only an early smoke test. Four local full-pulse attempts were deliberately interrupted at $t = 0.284454M$ after proving too costly; their partial histories do not establish a complete excursion.

8 Aurora PVC blocker

The isolated campaign root is

```
/lus/flare/projects/CompactBinaryMerger/hzhu/
  refgh_q_relaxed_20260826_e403baf_v1
```

Job 8785612 was a wrapper-only clean-tree failure and never built. Job 8785680 used e403baf0; job 8785718 used a3d98182. Both numerical jobs:

- mapped twelve distinct Aurora PVC tiles;
- configured Kokkos SERIAL;SYCL, Intel PVC, and GPU-aware MPI;
- built successfully with IntelLLVM 2025.3.2 and Kokkos 4.7.2;
- passed provider, reprojection, gauge, scalar-controller, and cache oracles on an actual PVC tile;
- completed eight-rank setup and wrote $t = 0$ histories;
- failed in the first evolved cycle on every rank with a Level Zero NotPresent write fault and exit status 134.

The intervening boundary refactor splits metric/gauge projection and maps one ghost cell per work item. Local pre/post histories are byte-identical. PVC spilling falls from about 119 registers to 19/17, but the fault remains on a different node. Therefore boundary spill pressure is not the established root cause. The combined q-shell reduction, repeated stage cache rebuild, and other first-cycle multi-rank paths remain hypotheses only. No third micro-experiment was submitted, and no owned Aurora job remains.

9 Phase audit and remaining work

Phase	Evidence
0	PASS : invalid same- r/h direct-FD gate is diagnostic-only.
1–5	PASS locally/source-unit: estimator, finite-radius oracles, weighted shell, finite- h target, and ϵ_G identity.
6–9	PARTIAL : provider and local metric/gauge/boundary algebra pass; evolved PVC path fails.
10–12	PARTIAL : controller and same-RK ordering implemented and local/manufactured gates pass; runtime restart is pending.
13	FAIL qualification: analytic jets and early smoke pass, but complete pulses and moving-reference convergence are absent; PVC first cycle fails.
14	PASS manufactured scalar histories; per-resolution runtime equilibration remains pending.
15	PARTIAL : regular annuli converge through $0.1M$; innermost metric Linf is nonmonotone.
16–18	Not run: closed loop, large mismatch, and three-resolution finite- h equilibrium.
19	PASS diagnostic policy; fixed- r/h nonconvergence retained.
20	PASS scope: uniform grid only; no SMR qualification attempted.
21	FAIL : evolved PVC and runtime restart/host/MPI/device matrix incomplete.

Before the controlling success phrase can be used, the first-cycle PVC fault must be fixed with a focused full-output gate. The four complete prescribed pulses, moving-q resolution ladder, closed-loop $q(0) = 0.9, 1.0, 1.1$ ladders, untuned 0.75/1.25 cases, finite- h equilibria, physical/constraint convergence, exact restart trajectory, and host/MPI/CUDA/PVC matrix must then all pass without intervention.

Compact logs, histories, convergence JSON, qstat records, mappings, provenance, and SHA-256 manifests live under `docs/fo_gh_artifacts/ref_gh_q_relaxed_controller_20260826`. Large field output and the 16 MB restart are intentionally excluded.